MIL-C-915/33B
23 August 1976
SUPERSEDING
MIL-C-915/33A
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE 6SGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

First - Copper conductor, uncoated, (see Details for size).
Second - Silicone rubber - glass tape insulation (see Details for diameter). Letter Identification Code, applied by method 5 (two conductors each of A, B and C).
Third - The six conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm, well-rounded assembly. Cabling sequence shall be ABC, ABC.
Fourth - Binder or combination binder/barrier.
Fifth - Special thermoplastic jacket (gray) (see Details for thickness). Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) The type designation shall change to 6SGA.
(b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(B) (e) The maximum overall dimension shall be increased by 0.050 inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage. When standard thermoplastic jacket is used, the air oven test does not apply.

Details

Type and size	Conductor size Navy standard	Diameter over insulation nominal (nom) (inch)	Jacket thickness nom (inch)	Overall diameter maximum (max) (inches)	Conductor resistance per 1000 ft at 25°Celsius (C) max (ohms)	Insulation resistance per 1000 ft minimum (min) (megohms)	Cold bending mandrel diameter max (inches)	Accelerated service loading (amps)
6SGU-100	100(61)	0.453	0.060	1.600	0.114	160	20	475
6SGU-125	125(61)	.507	.060	1.790	.0906	150	22	520
6SGU-150	150(61)	.557	.060	1.960	.0718	135	24	570
6SGU-200	200(61)	.634	.075	2.200	.057	125	27	650

(B) denotes changes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6145

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Voltage withstand - Volts, root mean square (Vrms), minimum (min)	5000
Insulation resistance - Megohms/1000 ft, min	(See Details)
Conductor resistance - Ohms/1000 ft, max	(See Details)
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Group B:	
Gas flame - 1 hour	No failure
Drip - At 95°C ±1°C	Zero
Cold bending, cable - (see Details for mandrel diameter)	No damage
Physicals (unaged)	
Jacket (Standard thermoplastic)	
Tensile strength - pound-force per square inch (lbf/in^2), min	1800
Elongation - Percent, min	225
Jacket (Special thermoplastic)	
(B) Tensile strength - lbf/in^2, min	2300
(B) Elongation - Percent, min	250
Group C:	
Physicals (aged)	
Jacket (Standard thermoplastic)	
Hot oil immersion	
Tensile strength - lbf/in^2, min	1400
Elongation - Percent, min	50
Jacket (special thermoplastic)	
(B) Air oven (168 hours at 136°C)	
Tensile strength - Percent of unaged, min	80
(B) Elongation - Percent of unaged, min	60
Hot oil immersion	
(B) Tensile strength - Percent of unaged, min	75
(B) Elongation - Percent of unaged, min	50
Permanence of printing (jacket) - Cycles, min	250
Cable filler removability	

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of group A, B and C plus the following:

Cable aging and compatibility (125°C ± 5°C)	No failure
Cold working (minus 20°C ± 2°C)	No damage
Gas flame (3 hours)	No failure
(B) Accelerated service - (All sizes) (See Details for load current)	No failure

UNIT ORDERING LENGTHS:
All sizes 800 feet (nom)

Custodians:
Army - MI
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH

(Project 6145-0674-33)

Review activities:
Army - EL, AV
Navy - EC

User activities:
Army - ME, MU, WC
Navy - CG

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